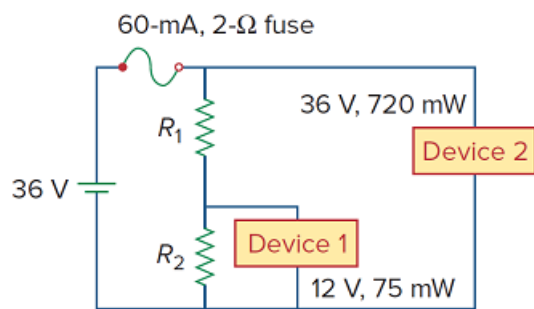


Problem

Two delicate devices are rated as shown in Fig. 2.142. Find the values of the resistors R_1 and R_2 needed to power the devices using a 36-V battery.

Figure 2.142:



Step-by-step solution

Step 1 of 4

Refer to circuit diagram in Figure 2.142 in the text book.

Calculate the current through device 2.

$$\begin{aligned} I_2 &= \frac{720 \text{ mW}}{36 \text{ V}} \\ &= 20 \text{ mA} \end{aligned}$$

Calculate the current through device 1.

$$\begin{aligned} I_1 &= \frac{75 \text{ mW}}{12 \text{ V}} \\ &= 6.25 \text{ mA} \end{aligned}$$

Step 2 of 4

Apply Kirchhoff's current law to calculate the current through the resistor R_1 .

$$\begin{aligned} 60 \text{ mA} &= I_{R_1} + I_2 \\ 60 \text{ mA} &= I_{R_1} + 20 \text{ mA} \\ I_{R_1} &= 60 \text{ mA} - 20 \text{ mA} \\ &= 40 \text{ mA} \end{aligned}$$

Similarly, calculate the current through the resistor R_2 .

$$\begin{aligned} I_{R_1} &= I_{R_2} + I_1 \\ 40 \text{ mA} &= I_{R_2} + 6.25 \text{ mA} \\ I_{R_2} &= 40 \text{ mA} - 6.25 \text{ mA} \\ &= 33.75 \text{ mA} \end{aligned}$$

Step 3 of 4

The voltage across the device 1 is,

$$V_1 = 12 \text{ V}$$

Calculate the value of resistor R_2 .

$$R_2 = \frac{V_1}{I_{R_2}}$$

Substitute 12 V for V_1 , and 33.75 mA for I_{R_2} .

$$\begin{aligned} R_2 &= \frac{12 \text{ V}}{33.75 \text{ mA}} \\ &= 355 \Omega \end{aligned}$$

Therefore, the value of resistor, R_2 is 355Ω .

Step 4 of 4

The voltage across the resistor R_1 is,

$$\begin{aligned} V_{R_1} &= V_2 - V_1 \\ &= 36 \text{ V} - 12 \text{ V} \\ &= 24 \text{ V} \end{aligned}$$

Calculate the value of resistor R_1 .

$$R_1 = \frac{V_{R_1}}{I_{R_1}}$$

Substitute 24 V for V_{R_1} , and 40 mA for I_{R_1} .

$$\begin{aligned} R_1 &= \frac{24 \text{ V}}{40 \text{ mA}} \\ &= 600 \Omega \end{aligned}$$

Therefore, the value of resistor, R_1 is 600Ω .